

CLASS TOPIC

Joint distributions, covariance and correlation

What this class fixes

- Know the central question of the topic.
- Use the correct notation and units.
- Avoid mechanical calculation without interpretation.

W06**Statistics**

Clean formulas · aligned layout · corrected te



CLASS TOPIC

Joint distributions, covariance and correlation

Economic question first

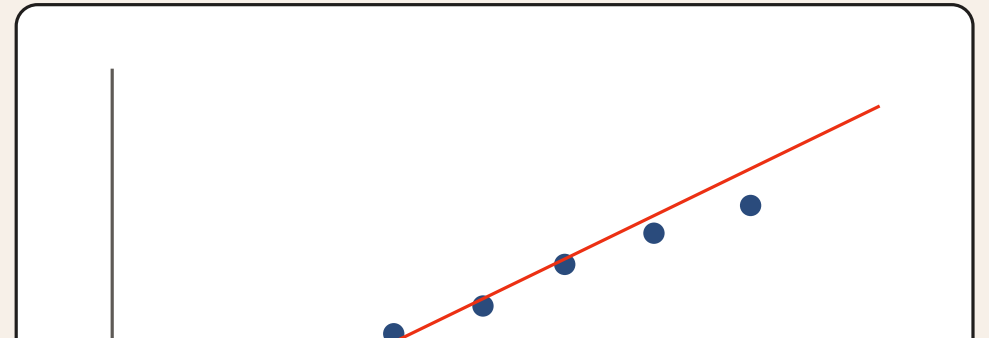
- Start from a real decision: compare, describe, forecast or test.
- Translate the question into variables before using formulas.
- Define the population, sample and measurement unit.

FORMULA

$$\text{Cov}(X,Y) = E[(X-\mu_x)(Y-\mu_y)]$$

sample cov =

$$\Sigma(x_i - \bar{x})(y_i - \bar{y}) / (n-1)$$



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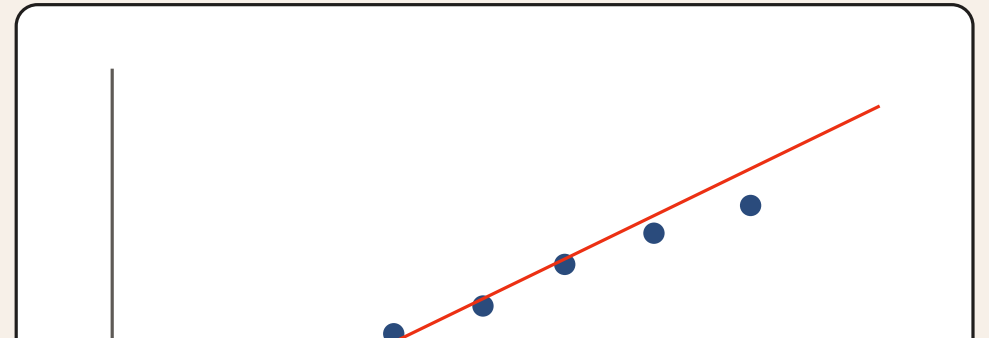
Joint distributions, covariance and correlation

Key definitions

- Population: full group of interest.
- Sample: observed part of the population.
- Statistic: number computed from a sample;
parameter: number for the population.

FORMULA

$$\text{Cov}(X,Y) = E[(X-\mu_x)(Y-\mu_y)]$$
$$\text{sample cov} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{(n-1)}$$



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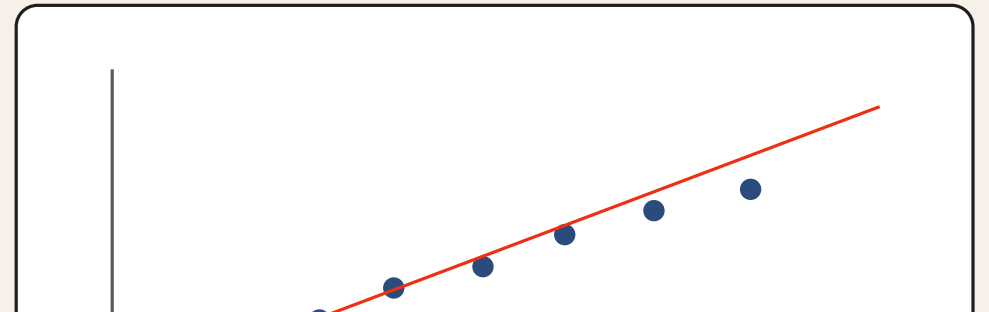
Joint distributions, covariance and correlation

Main formula box

- All formulas are kept in one merged formula area on this slide.
- Read every symbol before calculating.
- Use parentheses carefully.

FORMULA 1

$$\text{Cov}(X,Y) = E[(X-\mu_x)(Y-\mu_y)]$$
$$\text{sample cov} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{(n-1)}$$



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Mini example

- Write the data clearly.
- Substitute values step by step.
- Round only at the final reporting stage.

01 Data: 4, 7, 8, 9, 12

02 Step 1: compute the needed statistic

03 Step 2: write the result with units.

04 Step 3: explain it in one sentence.

FORMULA

$$\text{Cov}(X, Y) = E[(X - \mu_x)(Y - \mu_y)]$$

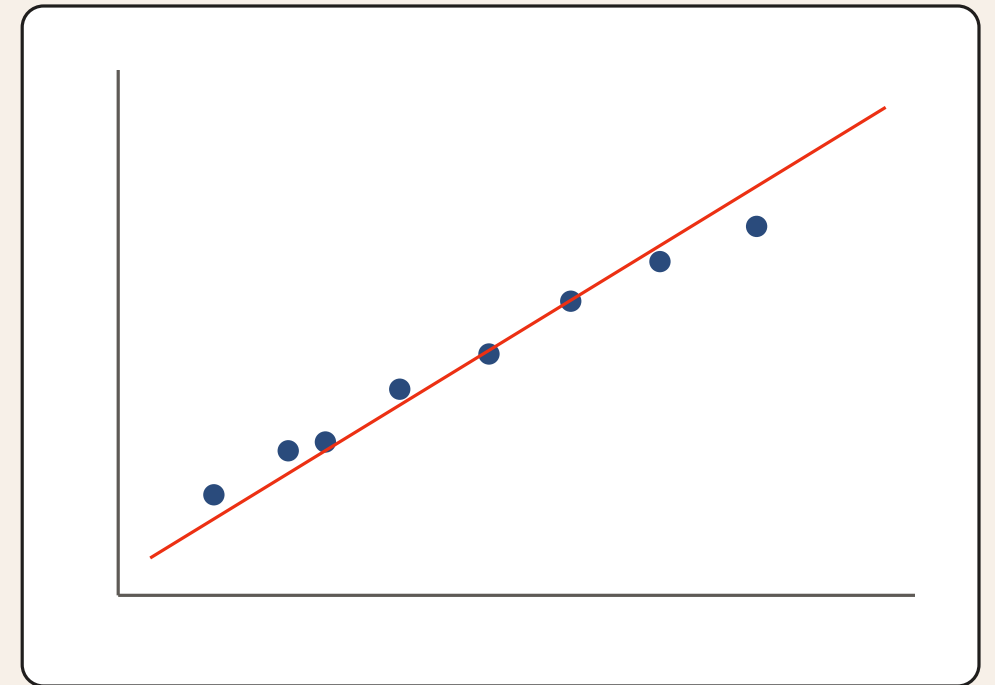
sample cov =

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Visual intuition

- A graph should answer a question, not decorate the slide.
- Always label axes and units.
- Look for center, spread, shape and unusual values.



FORMULA

$$\text{Cov}(X,Y) = E[(X-\mu_x)(Y-\mu_y)]$$

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Second formula box

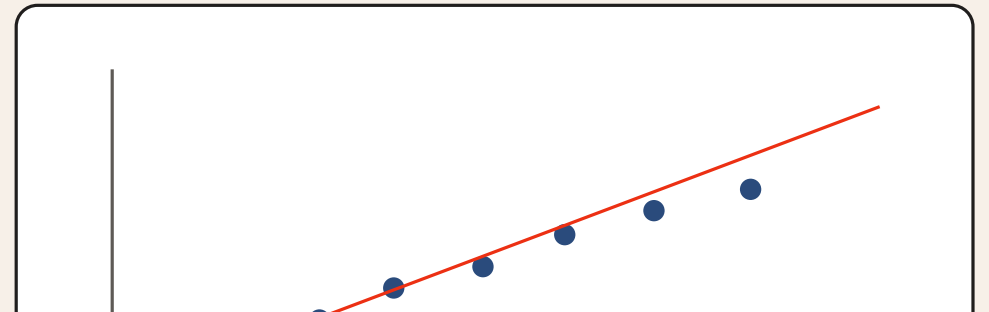
- This formula is separated from explanations but kept in one clean box.
- No broken equation fragments or floating text boxes.
- Notation is consistent across English and Russian versions.

FORMULA 2

$$\rho = \text{Cov}(X, Y) / (\sigma_x \sigma_y)$$

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{[\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2]}}$$

Formula boxes fixed: one merged box, consistent symbols, no broken fragments.



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Common mistakes

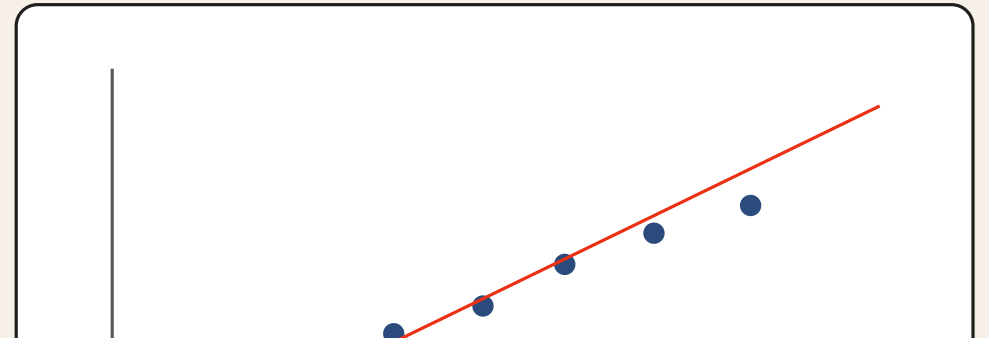
- Mixing sample and population symbols.
- Using a formula without checking conditions.
- Reporting a number without units or interpretation.

FORMULA

$$\text{Cov}(X,Y) = E[(X-\mu_x)(Y-\mu_y)]$$

sample cov =

$$\Sigma(x_i-\bar{x})(y_i-\bar{y})/(n-1)$$



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R / spreadsheet check

- Use software to calculate, but keep the statistical logic visible.
- Check variable names, missing values and units first.
- Reproducible work is easier to grade and correct.

R / EXCEL

```
# example pattern  
mean(x); sd(x); summary(x)  
plot(x, y); cor(x, y)
```

FORMULA
$$\text{Cov}(X, Y) = E[(X - \mu_x)(Y - \mu_y)]$$

sample cov =

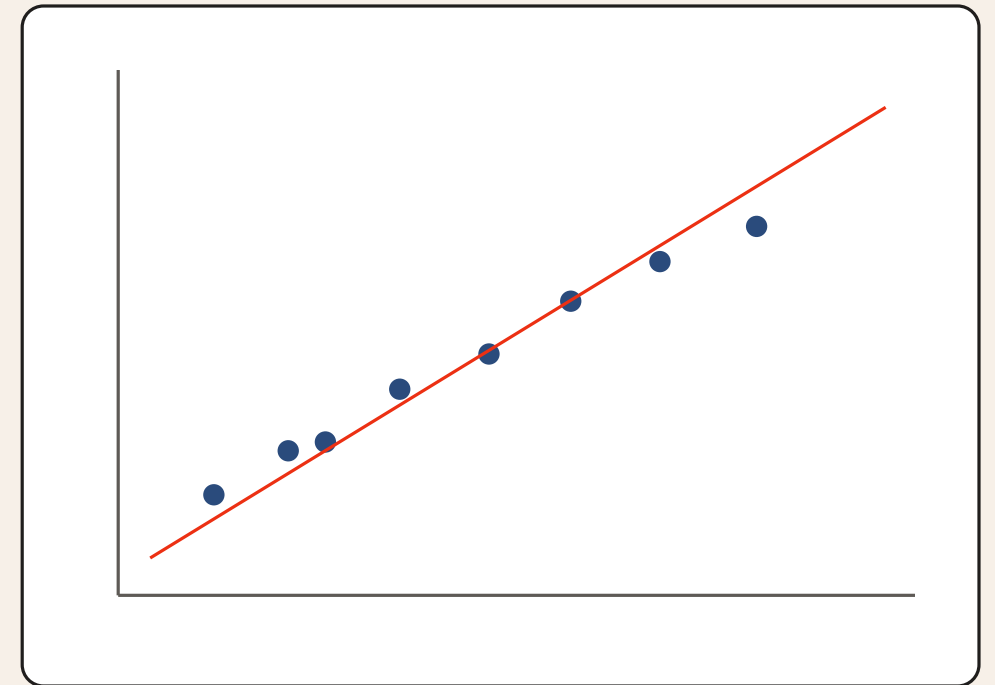
$$\sum (x_i - \bar{x})(y_i - \bar{y}) / (n - 1)$$

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Interpretation sentence

- Do not stop at the number.
- Explain what the result means in the original context.
- Use cautious language when evidence is limited.



FORMULA

$$\text{Cov}(X,Y) = E[(X-\mu_x)(Y-\mu_y)]$$

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Conditions and assumptions

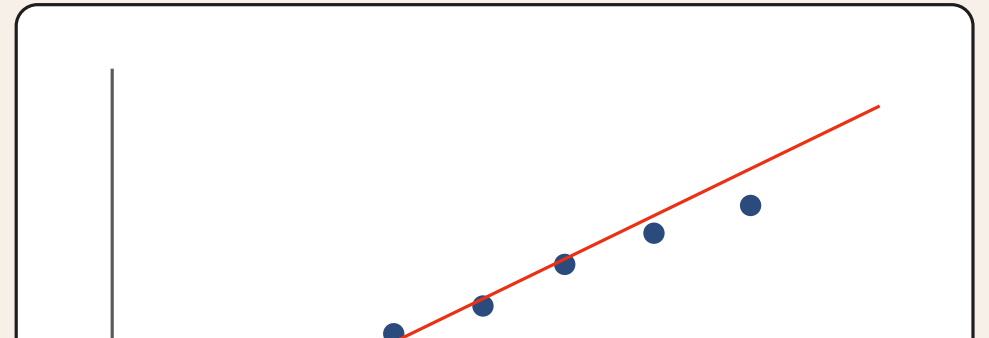
- Check whether the method fits the data type.
- Check independence and sample size when needed.
- If assumptions fail, explain the limitation.

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Class exercise

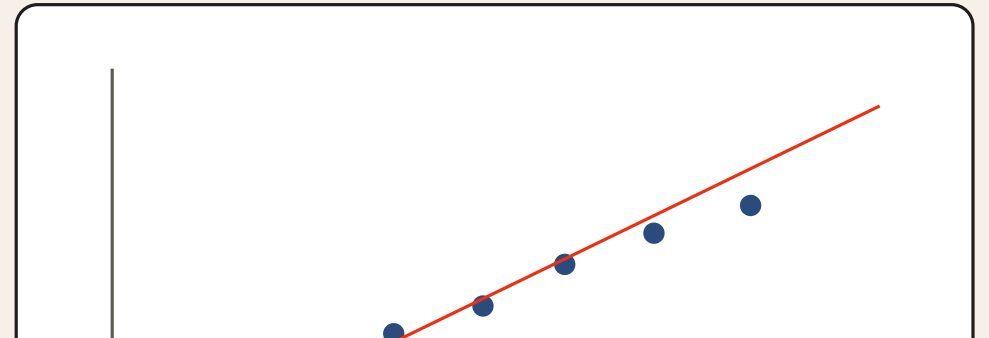
- Solve one calculation by hand.
- Then verify it with software.
- Write a two-sentence conclusion.

FORMULA

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sample cov =

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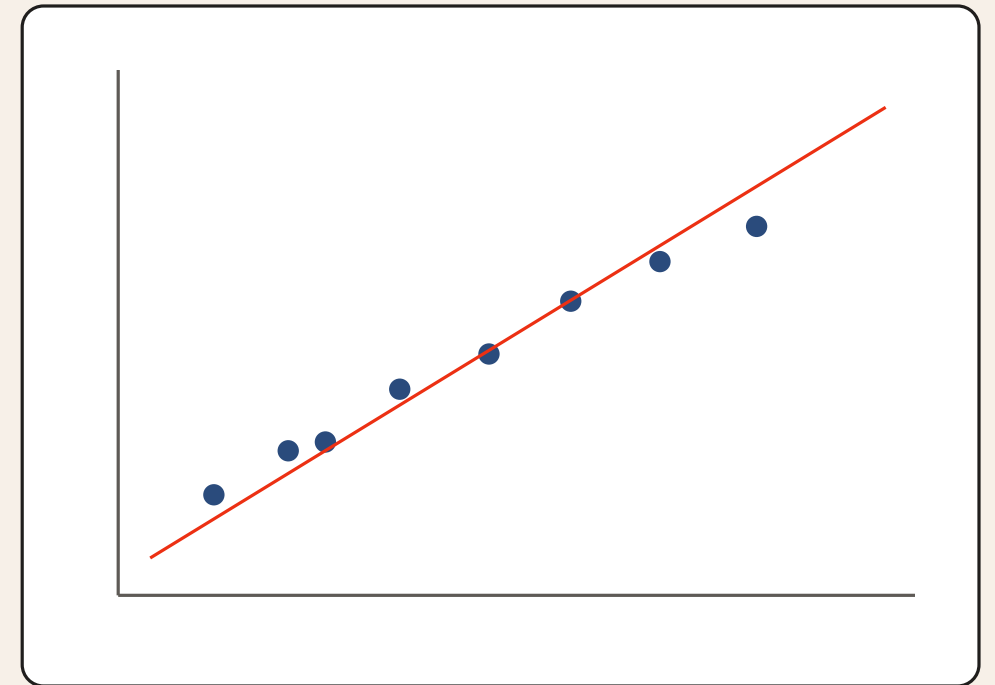


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How to report

- Report method, estimate, uncertainty and interpretation.
- Use tables only when they make comparison easier.
- Separate statistical evidence from causal claims.



FORMULA

$$\text{Cov}(X,Y) = E[(X-\mu_x)(Y-\mu_y)]$$

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Final checklist

- Question is clear.
- Variables and units are defined.
- Formula is correct.
- Result is interpreted.
- Limitations are stated.

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